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FEEDBACK LOOP ANALYSIS

Heavy Crane Rental Company — Quality Management

Farid Hajiyev • Ali Gasimov • Rovshan Guliyev • Elshad Samadov • Zaur Akbarov

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Verified by: Physics Teacher Azerbaijan Telman Askeraliyev (Fizika Muellimi, Azerbaijan, Baku)

01 — INTRODUCTION: FEEDBACK LOOPS IN QUALITY MANAGEMENT

What Are Feedback Loops?

A feedback loop is a process in which outputs of a system are routed back as inputs to the same system. In Quality Management, feedback loops are the engine behind continuous improvement — they transform raw operational data into corrective action.

KAIZEN CONNECTION

Kaizen (改善) demands small, continuous improvements. Feedback loops are the detection mechanism — they surface deviation before it becomes defect.

PDCA CYCLE

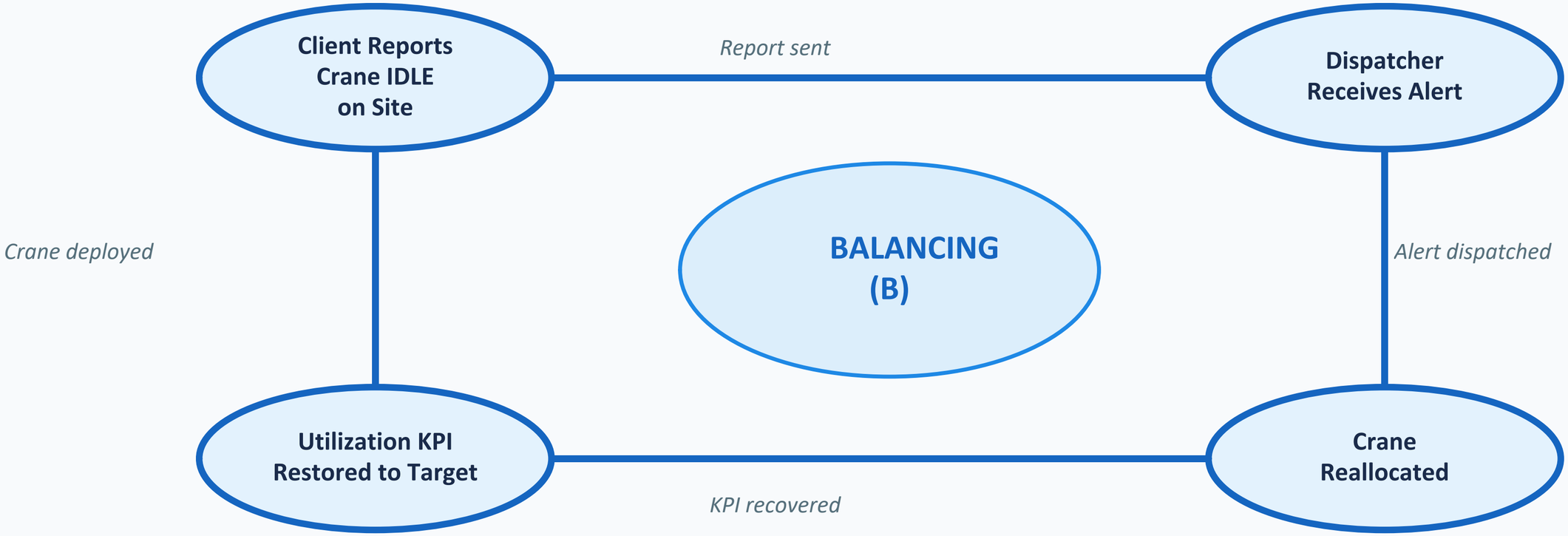
Plan → Do → Check → Act. The Check phase IS the feedback loop. Without data flowing back, PDCA stalls at 'Do' — execution without learning.

TWO LOOP TYPES

Balancing Loop
→ Seeks stability (thermostat model)
Reinforcing Loop
→ Amplifies change (virtuous or vicious)

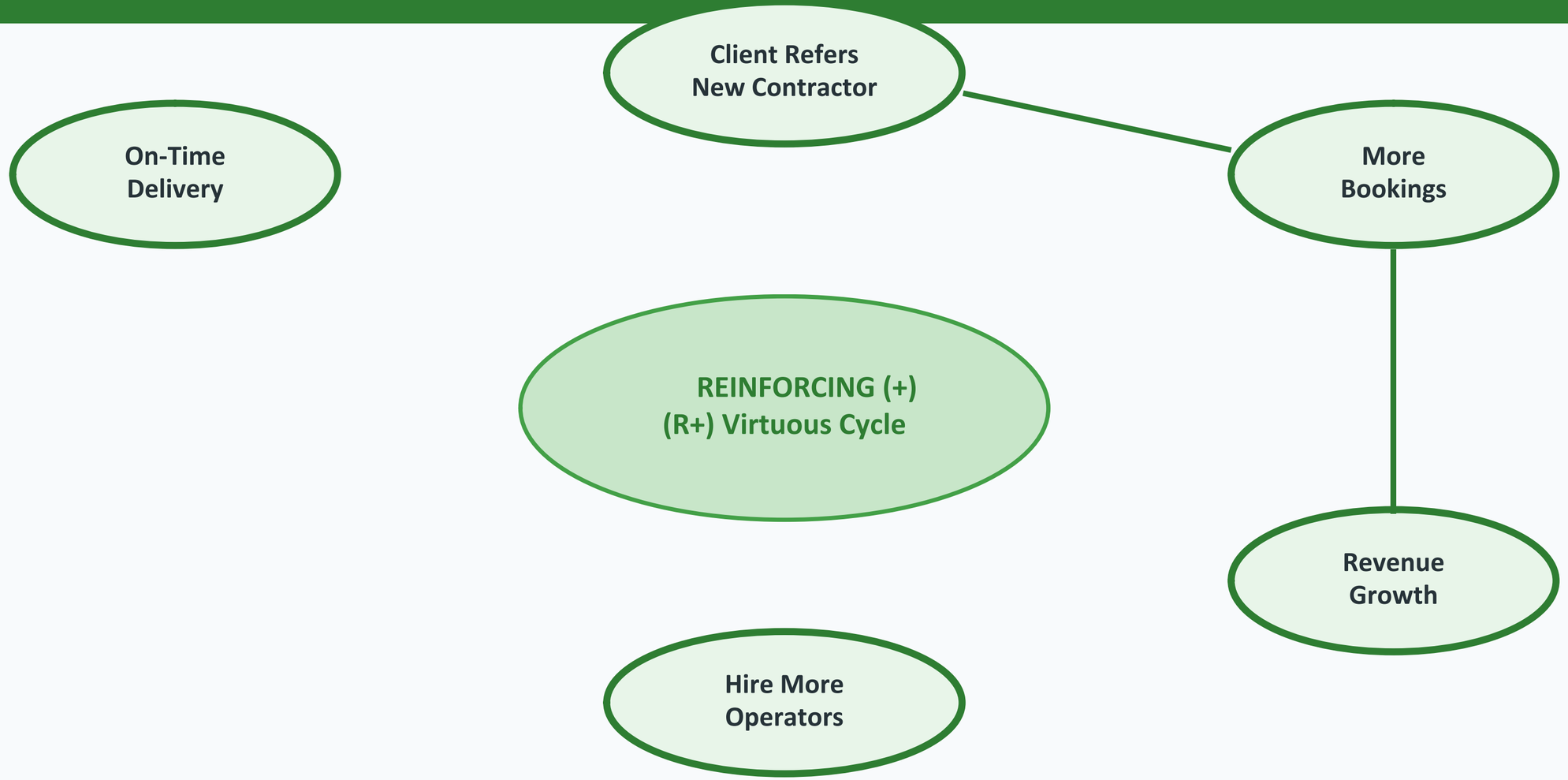
Applied Context: Heavy Crane Rental Company — where crane utilization, client satisfaction, and scheduling coordination create multiple interacting feedback dynamics.

02 — BALANCING LOOP: OPERATIONAL STABILITY



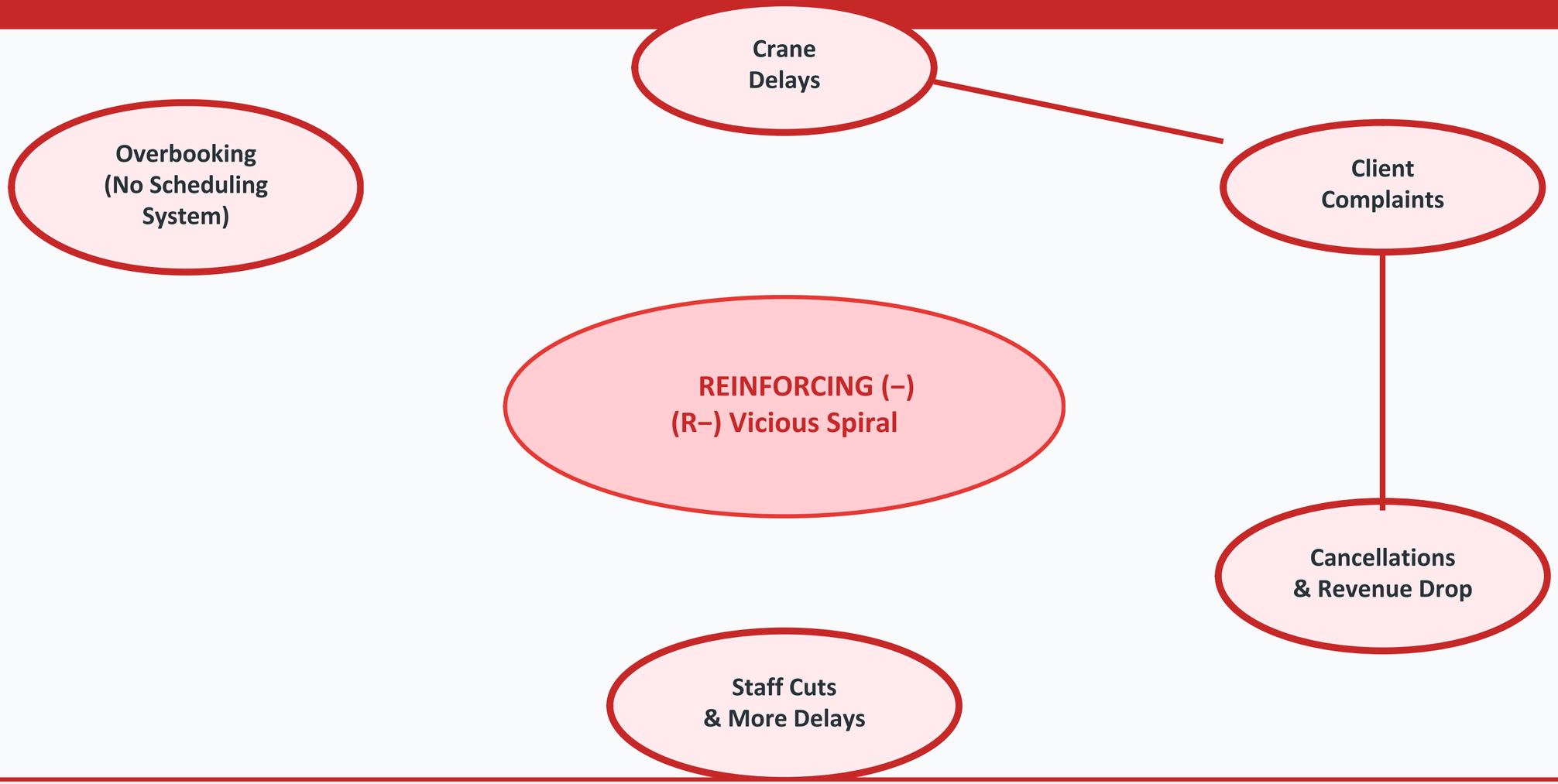
LESSON: The Balancing Loop acts as the operational thermostat — when crane utilization deviates from target KPI, the system self-corrects through dispatcher intervention. This is the stability engine of crane rental operations.

03 — REINFORCING LOOP (POSITIVE): THE VIRTUOUS CYCLE



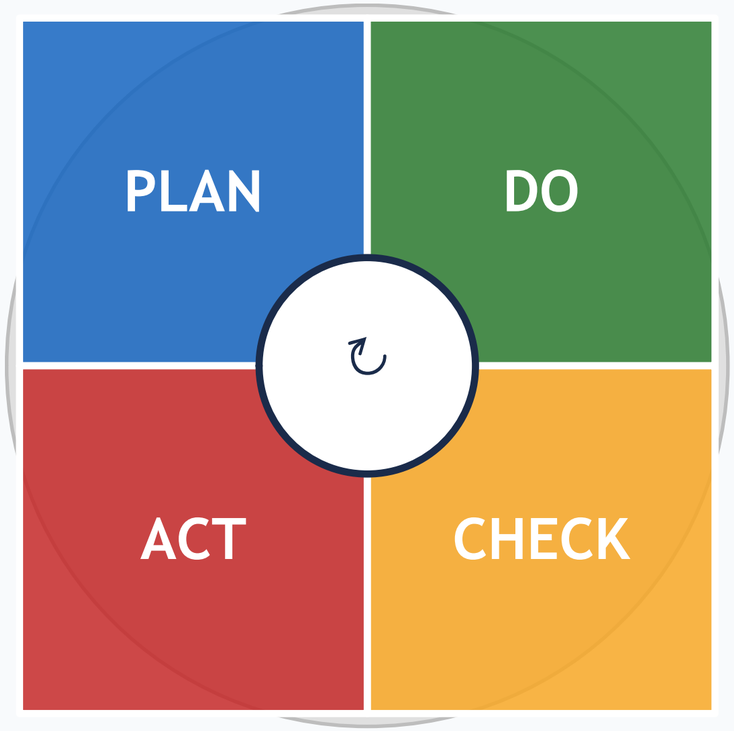
LESSON: This virtuous cycle must be actively managed — not taken for granted. Each node amplifies the next. Invest in on-time delivery systems and client relationship management to keep the cycle spinning upward.

04 — REINFORCING LOOP (NEGATIVE): COLLAPSE WARNING — MURI SPIRAL



MURI SPIRAL WARNING: Overburden in Japanese Lean (Muri) — the system collapses when scheduling constraints are ignored. Each failure triggers the next. A digital scheduling system breaks this loop before it starts.

05 — QM CONNECTION: PDCA CYCLE & FEEDBACK LOOPS



PLAN — Schedule crane allocation — reserve capacity per booking, assign operator rosters, define utilization targets.

DO — Execute mobilization — deploy cranes on-site, issue work orders, coordinate logistics.

CHECK — Feedback loop triggers — monitor utilization KPI, log client complaints, flag idle-time alerts. This IS the feedback.

ACT — Dispatcher intervention — reallocate resources, renegotiate contracts, update scheduling system rules.

Key Insight: Without the CHECK phase generating feedback, the PDCA loop breaks. Feedback loops ARE the measurement infrastructure that makes PDCA work.

06 & 07 — CROSS-DOCUMENT CONNECTIONS: FISHBONE + SIPOC

FILE 1: FISHBONE DIAGRAM LINK

Root Cause Identified:

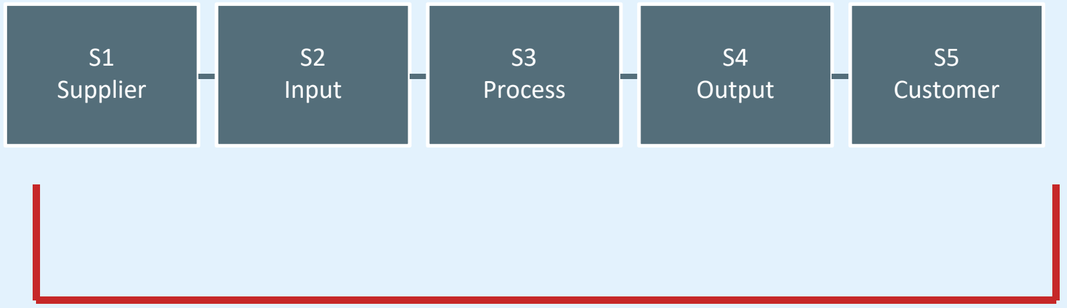
"Weak Scheduling & Coordination"

How it breaks the Balancing Loop:

→ Dispatcher cannot reallocate cranes if scheduling data is absent or inaccurate
→ The self-correction mechanism fails — the thermostat has no sensor
→ Balancing loop converts to Reinforcing Negative spiral (Muri)

FILE 2: SIPOC MAP LINK

Missing Feedback Mechanism:



NO FEEDBACK FLOWS BACK

Gap: No data flows from Step 5 (Execution) back to Step 1 (Inquiry & Quote)

Impact: Historical demand signals, client satisfaction scores, and utilization data never inform future quoting — the cycle is open-loop

08 — PROPOSED SOLUTION: CLOSING THE FEEDBACK LOOP

SOLUTION 1: Real-Time KPI Dashboard

Closes the Feedback Loop Automatically

- Live crane utilization rate — updated every 15 min
- Idle time alert → auto-notifies dispatcher
- Client satisfaction score from post-job survey
- Booking-to-deployment lead time tracker
- Connects Step 5 (Execution) back to Step 1 (Inquiry)

SOLUTION 2: Weekly PDCA Review Meeting

Institutionalizes the Check-Phase

- Monday review: Check-phase data from prior week
- Act phase: dispatcher rule updates, contract renegotiation
- Cross-team: Sales, Operations, Maintenance aligned
- Kaizen log: capture micro-improvements each cycle
- KPI trend shared with all authors & management

Both solutions together close the open loop identified in the SIPOC Map — transforming operations from reactive to predictive quality management.

SUMMARY: THREE FEEDBACK LOOPS AT A GLANCE

BALANCING LOOP

Stability

Idle crane → Dispatcher alert → Reallocation → KPI restored. The operational thermostat. Broken by weak scheduling (Fishbone Root Cause).

REINFORCING (+)

Growth

On-time delivery → Referrals → Bookings → Revenue → Hire more operators. Virtuous cycle requiring active management and client-first culture.

REINFORCING (–)

Collapse

Overbooking → Delays → Complaints → Cancellations → Staff cuts → More delays. Muri spiral triggered by absent scheduling system.

PDCA Integration: Plan (Schedule) → Do (Deploy) → Check (KPI Dashboard) → Act (Dispatcher / PDCA Review Meeting)

Thank You

Heavy Crane Rental Company — Feedback Loop Analysis

Authors:

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